EMERGE HACKATHON · SYLVESTER HEALTHCARE TRACK

BUILT IN ONE DAY

Survivorship 360

One place to see who needs them today, and why.

Four AI tools for cancer survivorship, in one care-team screen. Ask it in plain words from ChatGPT or Claude.

patient-360.raava.dev

THE PROBLEM

Too many signals. No ranking.

WEARABLE + CHECK-INS

Who is heading into a rough week?

SURVEYS + CARE USE

Who may end up in the ER next year?

GLUCOSE SENSOR

Which daily habit moves their blood sugar?

COACHING CALLS

Did they just say something urgent?

Today, the coach hears about a bad week at the next call, days late.

THE PRODUCT

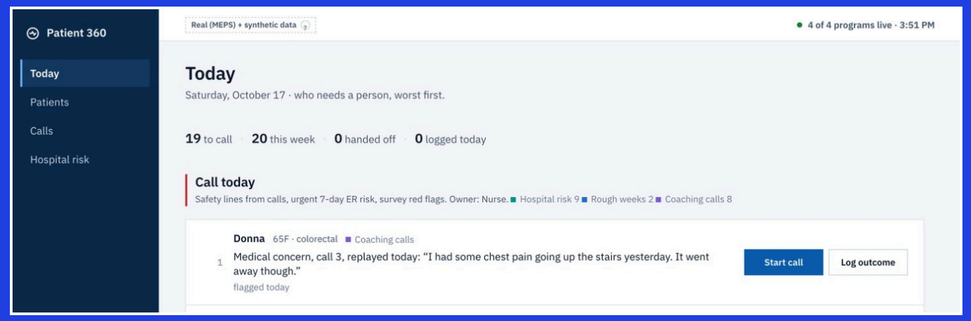
Two front doors, one set of answers.

FOR THE CARE TEAM

Patient 360

A Today list of who to call and why. A 360 view per survivor. Live coaching calls with a recap. An AI assistant that answers and navigates.

`patient-360.raava.dev`



FOR ANY AI ASSISTANT

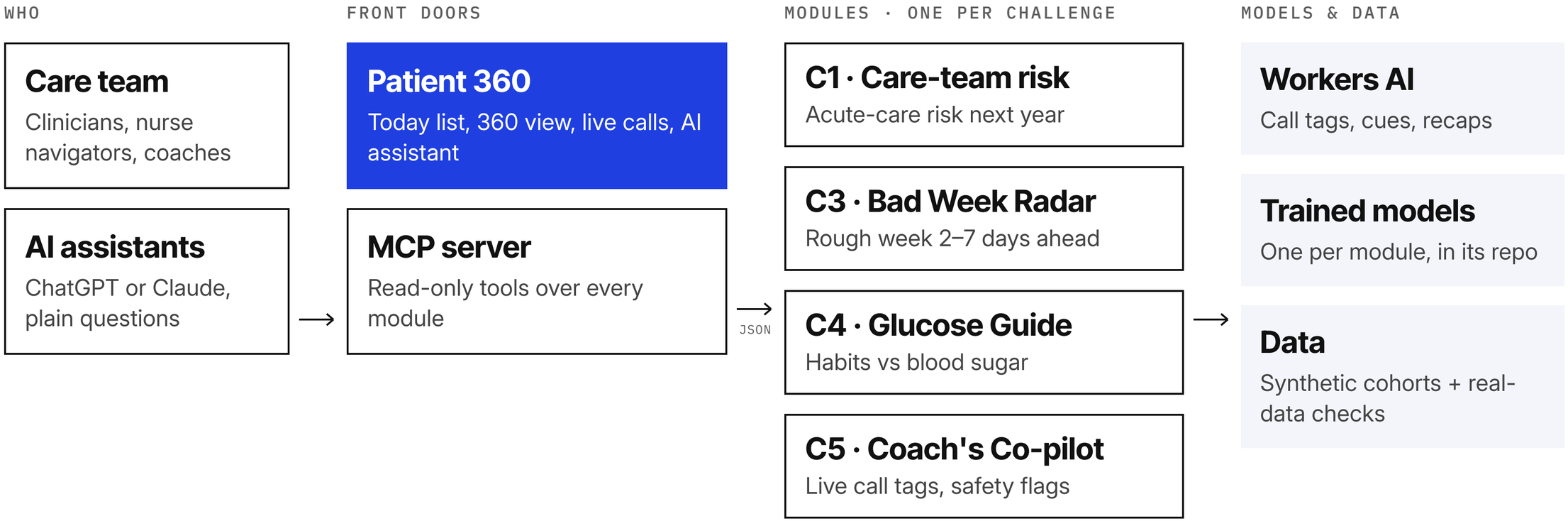
MCP server

12 read-only tools for ChatGPT and Claude: "who should we call today?", one survivor across modules, "score this conversation".

`emerge-mcp.raava.dev/mcp`

ARCHITECTURE · ALL ON CLOUDFLARE WORKERS

One contract per module, two ways in.



C1 · CARE-TEAM RISK

Who may need acute care next year?

HOW IT WORKS

Data

Real MEPS survey answers from 466 cancer survivors; 134 had an ER visit or hospital stay.

Model

Calibrated random forest, picked from 48 candidates, with each survivor's drivers.

Validation

Nested CV over 10 outer folds: no score saw its own test fold.

PROBABILITIES SIGNIFICANTLY BETTER THAN THE ORGANISERS' BASELINE

59%

of "50%+ risk" calls right [46%, 71%], vs 48% for the organisers' model

1.10

calibration slope (1 = ideal) vs 0.45

0.539

log loss vs 0.606, a significant gain

Out-of-fold results. Ranking: AUROC 0.711 vs 0.672, better but not significant.

HONEST LIMIT

It ranks NH Black survivors poorly (AUROC 0.48, n = 47), and the panel warns. Prior care use dominates: it marks need, not a lever.

C3 · BAD WEEK RADAR

Who is heading into a rough week?

HOW IT WORKS

Data

Their own check-ins against their usual, skipped check-ins, the wearable and the treatment calendar.

Model

Gradient boosting forecasts a rough patch within 7 days. A second model flags likely ER days.

Validation

750 patients, split by patient. Every method gets the same budget: about one new alert per coach per day.

RESULTS · HELD-OUT PATIENTS, VS CALENDAR BASELINE

48%

of rough patches caught 2–7 days early, vs 36% for the calendar baseline

47%

of ER visits caught 2–7 days early, vs 36% for a calendar-only ER model

18%

of alerts followed by a rough patch, vs 11%

Lift over the calendar: +8 to +15 points of patches caught (95% CI). A nudge to message or call earlier, not a diagnosis.

Adding a model-free escalation (red-flag diary notes, passive suicidal ideation in English, Spanish or Creole, a day rated 1 of 5): 61% vs 47% (CI +9 to +19), for about 4 more calls per coach per week.

HONEST LIMIT

In a real fatigue study the wearable does not forecast the next day (AUC 0.482), so lead time is shown on synthetic data. On later dates (trained before July 2025, tested from October 2025), 44% vs 36%, CI −1 to +17: same direction, not significant. Only sarcoma is clearly below the overall recall: 13% vs 28%.

C4 · GLUCOSE GUIDE

Which habits move their blood sugar?

HOW IT WORKS

Data

500 sensor wearers and 18,218 meal curves from a synthetic 12-week program.

Model

Double ML within person: each survivor against themselves. Personal advice only where their own data supports it.

Validation

Cross-fitted. The walk effect passes a negative control; every claim carries a real-data check label.

RESULTS · REAL ADULTS FIRST · WITHIN PERSON, 95% CI

−3.2

mg/dL, walk after a meal, 44 real adults (CGMacros): not distinguishable from zero

−18.3

mg/dL, walk after dinner, synthetic program (CI 16.6–20.1)

+9.2

mg/dL when dinner is at 8 pm or later (7.1 to 11.3)

In 44 real adults (CGMacros), a walk after a meal went with a rise only 3.2 mg/dL smaller (95% CI −6.9 to +0.5): not distinguishable from zero. In the synthetic program, a walk after dinner shows −18.3 mg/dL (CI 16.6–20.1).

HONEST LIMIT

Walks were self-chosen, so this is an association. The real interval includes zero: expect a much smaller effect in real people than in the program.

C5 · COACH'S CO-PILOT

How is the coaching call going?

HOW IT WORKS

Data

1,085 synthetic coaching calls in English and Spanish, split by participant.

Model

Cloudflare Clef tags every line with an MI code: 0.3–0.5 s model time per line. GLM writes a coaching cue only when one is needed.

Validation

Tuned on 20% of participants. The other 80% held out and reported only.

RESULTS · HELD-OUT CALLS (56,001 ANSWERED LINES)

r 0.75

agrees with each supervisor about as well as they agree with each other: r 0.75 with A, 0.70 with B, vs 0.69

93.1%

of coach moves tagged right (12 labels)

13/13

urgent lines caught. Suicide warning signs: 44/54 on an independent held-out set (was 25/54).

Cost: 244 false alarms in 25,785 lines (was 9), about 1 call in 5. Red banners for medical, distress and social needs; amber when the coach misses one or gives medical advice.

HONEST LIMIT

Still misses about 1 in 5 indirect warning signs, so every distress flag tells the coach: use your suicide-risk screening protocol now. Haitian Creole is tested only on the safety flag. Never tuned on AnnoMI: it agrees 75.0% (experts 83.9%).

HOW TO READ OUR NUMBERS

Every number traces to a validation run.

Split by person, scored once

Held-out patients for C3 and C5, nested out-of-fold CV for C1, cross-fitting and falsification checks for C4.

A baseline next to every headline

The organisers' model, the treatment calendar, or expert agreement. Small gains are called small.

Weak spots shown in the product

Fairness gaps and real-data checks appear in the UI, not only in the README.

WHAT WE TELL JUDGES UP FRONT

C1	Ranks NH Black survivors poorly: AUROC 0.48, n = 47
C3	Real wearable does not forecast the next day: AUC 0.482
C4	44 real adults: walk −3.2 mg/dL, not distinguishable from zero (program −18.3)
C5	Agrees with real experts 75.0%; experts agree 83.9%

The module cohorts are separate synthetic datasets that share no patient ids. Cross-module views use composite personas, labelled as such.

LIVE DEMO

Five steps, three minutes.

1 **The Today list**
Who the care team should call today, across modules, and why.

2 **One survivor's 360 view**
Risk, week outlook, glucose advice and coaching in one place.

3 **A coaching call, live**
Tags, fidelity score and safety banners appear line by line.

4 **Ask the AI assistant**
The chat assistant answers and takes you to the right screen.

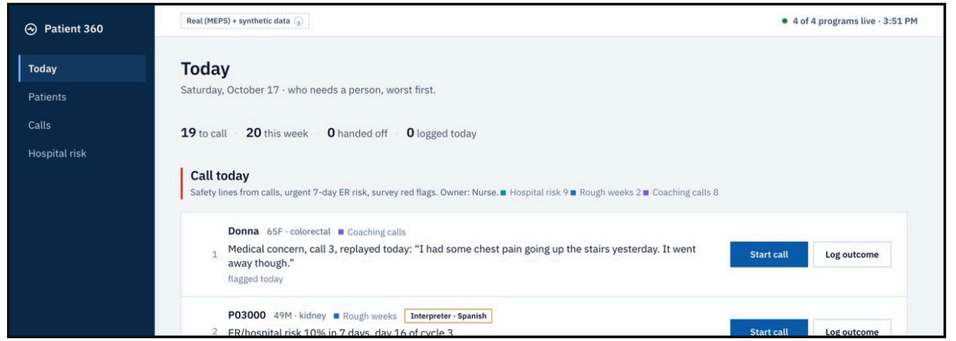
5 **Ask Claude or ChatGPT**
Through the MCP server: "who should we call today?"

PATIENT 360
patient-360.raava.dev

MCP SERVER
emerge-mcp.raava.dev/mcp

C5 COACH'S CO-PILOT
coach.raava.dev

GITHUB MONOREPO
github.com/zaycruz/survivorship-360



WHAT'S NEXT

From a one-day build to real survivors.

C1

Predict first events

Not last year's repeat users. Prevention needs the first event, and no model finds it well yet.

C3

Use yesterday's rating

A causal scoring mask unlocks the strongest next-day signal in real data.

ALL MODULES

Validate on real data

Re-check every synthetic effect against real survivorship records.

OUR PROPOSAL · TEAM 1

A pilot with the Sylvester survivorship care team.

github.com/zaycruz/survivorship-360

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